

int[] arr = {64, 34, 25, 12, 22, 11, 90};

mergeSort(arr, 0, 6) → [64, 34, 25, 12, 22, 11, 90] → mid = 3

Split into:

Left → (0–3) → [64, 34, 25, 12]

Right → (4–6) → [22, 11, 90]

LEFT HALF (0–3)

mergeSort(arr, 0, 3) → mid = 1

Split into:

Left → (0–1) → [64, 34]

Right → (2–3) → [25, 12]

LEFT of LEFT (0–1)

mergeSort(arr, 0, 1) → mid = 0

Split into:

- (0–0) → [64]
- (1–1) → [34]

Merge [64] and [34] → [64,34] → [34,64] =====> [34, 64, 25, 12, 22, 11, 90]

RIGHT of LEFT (2–3)

mergeSort(arr, 2, 3) → mid = 2

Split:

- (2–2) → [25]
- (3–3) → [12]

Merge [25] and [12] → [25,12] → [12,25] =====> [34, 64, 12, 25, 22, 11, 90]

Finally, Merge (0–1) and (2–3)

[34, 64] and [12, 25] $\Rightarrow$ [12, 25, 34, 64] $\Rightarrow$ [12, 25, 34, 64, 22, 11, 90]

RIGHT HALF (4–6)

mergeSort(arr, 4, 6) $\Rightarrow$ mid = 5

Split:

- (4–5) $\rightarrow$ [22, 11]
- (6–6) $\rightarrow$ [90]

LEFT of RIGHT (4–5) $\Rightarrow$ mid = 4

Split:

- [22]
- [11]

Merge [22] and [11] $\Rightarrow$ [22, 11] $\Rightarrow$ [11, 22] $\Rightarrow$ [12, 25, 34, 64, 11, 22, 90]

Merge (4–5) and (6–6)

[11, 22] and [90] $\Rightarrow$ [11, 22, 90] $\Rightarrow$ [12, 25, 34, 64, 11, 22, 90]

Finally merging for left and right array, [12, 25, 34, 64] and [11, 22, 90]

[12, 25, 34, 64, 11, 22, 90] $\Rightarrow$ [11, 12, 22, 25, 34, 64, 90]

